



New Archaeological Examples of Dental Prosthetics: Dentures from the Nineteenth Century Village of Middenbeemster, The Netherlands

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Introduction & Cases

Archaeological examples of dental prosthetics provide insight about past cultural practices, technological development, and oftentimes provenience (e.g. 1, 2, 3, 4, 5).

In the post-Medieval rural village of Middenbeemster (fig. 1), The Netherlands, two individuals with dentures were uncovered from the a churchyard containing approximately 450 skeletons. Both individuals lived in the 1800's. Individual S404 is a 26-35 year-old male with a small inset of three fake teeth amongst the mandibular incisors (figs. 2a and b). Calculus deposits had begun to accumulate on the lingual surface, and a smooth underlying alveolus, suggest the prosthetic had been worn for some time.



Fig. 1. Middenbeemster, The Netherlands



Figs. 2a and b: Individual S404, young adult male, with carved bone teeth in location of LI₁, RI₁ and RI₂. Tiny drilled holes run antero-posteriorly on the inferior outer edge from where the prosthetic was anchored to the teeth. Note large pipe-notches between the maxillary I²-Cs.



Archival records identify individual S914 as a 43 year-old female named Beletje W. She has large metal and porcelain-glass prosthetics spanning the anterior maxilla and mandible (figs. 3a and b). This individual had extensive antemortem tooth loss and periodontal disease, as well as severe caries in most of the few remaining teeth. The presence of concha bullosa, and active and healed lesions in the lacrimal/orbital region, suggest maxillary sinusitis (infection).



Figs. 3a & b. Individual S914, a 43 year-old female. In the maxilla the prosthetic runs from the LC to the RM¹ with the brace anchored around the remaining LPM¹, RPM¹ and RM². A tree root runs through the hole in the brace for the RPM¹. The tooth roots of the LC, LI² and RI² are visible. In the mandible the prosthetic runs from the LM₂ to the RPM₂ with the brace anchored around the remaining LPM₁.



Compositional Analysis

In individual S404 the fake teeth were examined under a stereoscopic microscope at 10x and 40x magnification. In individual S914 hand-held X-ray fluorescence (HH-XRF) was used to identify the material composition of the denture base and fake teeth. The Bruker Tracer III-SD HH-XRF was used without a filter on settings optimized to 40kv and 12µA with a vacuum system for detection of light elements. This setup excites all elements from Mg to Pu. Quantification estimate precision is +/-5%.

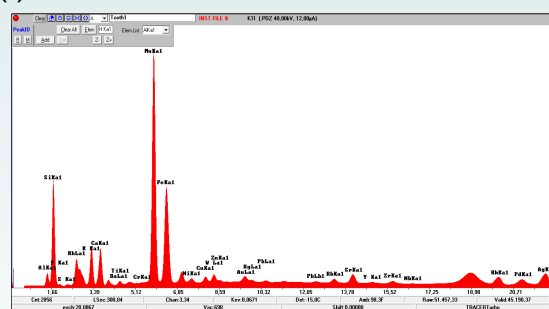
Results

Microscopic analysis of S404's denture revealed small areas of trabeculae proving the teeth are carved from bone. It is not possible to identify species at this time. In individual S914 two HH-XRF analyses showed the denture base is made of a silver copper alloy, with small amounts of zinc, gold, lead, and nickel (table 1). Based on the quantitative levels of elements (fig. 4) in S914's dentures they were made out of porcelain. The abundant presence of manganese (and trace amounts of tungsten) are however anomalous to tableware porcelain (6). The exterior of the denture is coated in a glass matrix. The manganese may have been used to counteract coloration providing a colourless or pale yellow result (7).

Table 1. HH-XRF quantitative compositional data from two locations on the denture base

		Denture01	Denture02
Ag	wt.%	85.28	89.29
Cu	wt.%	6.29	6.34
Zn	wt.%	2.35	0.66
Au	wt.%	1.39	0.80
Pb	wt.%	1.12	0.84
Ni	wt.%	3.57	2.07

Fig. 4. XRF analysis of teeth. The presence of K lines of Al, Si, P, S, K, Ca, Ba, Ti, Cr, Mn, Fe, Ni, Cu, Zn, Rb, Sr, Y, Zr, Nb, Rh, Pd & Ag and L lines of Rh, W, Au, Hg & Pb are consistent throughout the object.



Discussion & Conclusion

Traces of lead in the silver alloy point to a fabrication date prior to electrolytic refining in the 19th century (8). The copper-depleted/silver-enriched composition of the brace is consistent with 18th and 19th century alloy standards and correlates more with German, Austrian, or Italian standards rather than a British 'Sterling' or 'Britannia' standard (9). These interesting examples add to our understanding of past dental practices in mainland Europe and suggest that by the nineteenth century such novel and likely expensive devices were no longer limited to the very upper class.

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